Math 241, F04 **Name:**

Quiz 1 **SSN: xxx-xx-**

<u>Instructions:</u> **There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.**

Problem 1. (10 points) Given parametric equations

$$x = 3t^2,\ y = 4t^3; \quad t \neq 0.$$

Without eliminating the parameter,

a) Find dy/dx.

Solution:

$$\frac{dx}{dt} = 6t$$

$$\frac{dy}{dt} = 12t^2$$

$$y' = \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{12t^2}{6t} = 2t$$

b) Find d^2y/dx^2.

Solution:

$$\frac{dx}{dt} = 6t$$

$$\frac{dy'}{dt} = 2$$

$$\frac{d^2y}{dx^2} = \frac{dy'/dt}{dx/dt} = \frac{2}{6t} = \frac{1}{3t}$$

Problem 2. (10 points) Given two vectors

$$\mathbf{a} =< 1, -3 >,\ \mathbf{b} =< -1, 2 >,$$

a) Find $|\mathbf{a}|$ and $|\mathbf{b}|$.

Solution:

$$|\mathbf{a}| = \sqrt{1^2 + (-3)^2} = \sqrt{10}$$
$$|\mathbf{b}| = \sqrt{(-1)^2 + 2^2} = \sqrt{5}$$

b) Find $\mathbf{a} \cdot \mathbf{b}$.

Solution:

$$\mathbf{a} \cdot \mathbf{b} = 1 \times (-1) + (-3) \times 2 = -7$$

c) Find the cosine of the angle between $\mathbf{a}$ and $\mathbf{b}$.

Solution:

$$\cos\theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}|\ |\mathbf{b}|} = \frac{-7}{\sqrt{10}\sqrt{5}} = \frac{-7}{5\sqrt{2}}$$